MIE346: Design Assignments 1-4

Group #17

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1.0 Engineering Specification and Background Research

We are tasked with designing a circuit which can convert the input voltage signal (0 - 0.35 V) to the desired output voltage signal (0.25 - 3.3 V). This relationship can be expressed as:

$$V_{out} = V_{in} \times \frac{3.3 - 0.25}{0.35} + 0.25$$

where we have a positive gain of $\frac{61}{7}$ and a positive offset 0.25.

After conducting research, relevant circuits include: Basic Amplifier (Inverting and Non-Inverting), Summing Amplifier (Inverting and Non-Inverting), Difference Amplifier (Inverting and Non-Inverting), Buffer (Inverting and Non-Inverting), and the Two Resistor Divider. The decision was to select a single Non-Inverting Amplifier and a Difference Amplifier as the solutions to this design problem.

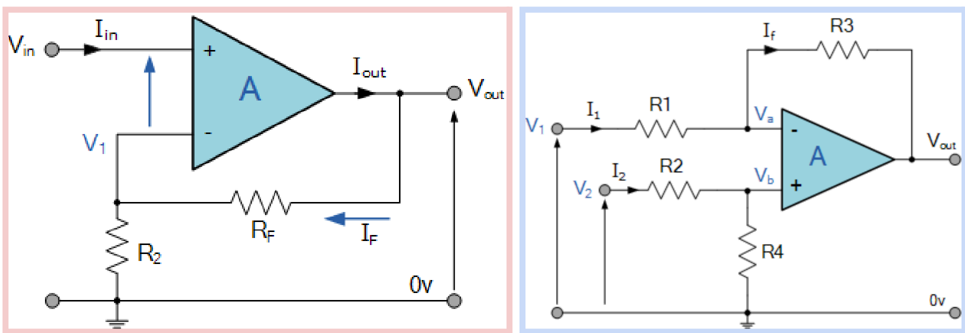

Figure 1: Non-Inverting Operational Amplifier (Red) [1] and Difference Amplifier (Blue) [2]

Upon analyzing the Non-Inverting Operational Amplifier [1], the relationship between V_{out} and V_{in} can be expressed as:

$$V_{out} = \left(1 + \frac{R_F}{R_2}\right) V_{In} .$$

Core Functionality: This Non-Inverting Operational Amplifier circuit can amplify the input voltage without changing its sign with the positive gain.

Weaknesses and Limitations: Comparing the relationship between V_{out} and V_{in} and the one we want to have, $V_{OUT} = V_{IN} \times \frac{3.3 - 0.25}{0.35} + 0.25$, We can find that the positive offset is 1 instead of 0.25. This offset directly depends on V_{In}. However, the component and overall circuit can be reused and modified to meet our needs and get the gain expression we want. The design uses an additional DC voltage source which needs to be modified in order to meet the engineering specification detailed in the assignment, as we are not allowed to use additional voltage sources for our circuit design.

The second circuit, the Difference Amplifier, was also chosen for its usefulness towards the design project. Upon researching the circuit [2], the expression between V_{out} , V_1 and V_2 is

$$V_{out} = \frac{R_3}{R_1}(V_2 - V_1) \text{ and } \frac{R_3}{R_1} = \frac{R_4}{R_2}.$$

Let us assign V2 as V_{IN} and V_1 as an additional voltage source. This expression is very similar to the one we want: $V_{out} = V_{in} \times \frac{3.3 - 0.25}{0.35} + 0.25$ if we make $\frac{R_3}{R_1} = \frac{3.3 - 0.25}{0.35}$ and $V_1 = -0.25 \times \frac{0.35}{3.3 - 0.25}$.

Core Functionality: The difference between input voltages is linearly proportional to the difference between output voltages. That is the output voltage if we measure the voltage relative to the ground. Additionally, the gain of this circuit is not affected by the non-zero load resistor connected across the output side of the amplifier.

Weaknesses and Limitations: The circuit is relatively more complicated as it requires more components, thus also leading to an increase in cost. It is also important to note that a more complicated circuit has a higher possibility of failure. Any one component could jeopardize the entire circuit, needing further investigation and time spent. This design also needs to be modified as it uses an additional DC voltage source.

2.0 Candidate Designs

2.1 Candidate Design 1: Non-Inverting Operational Amplifier

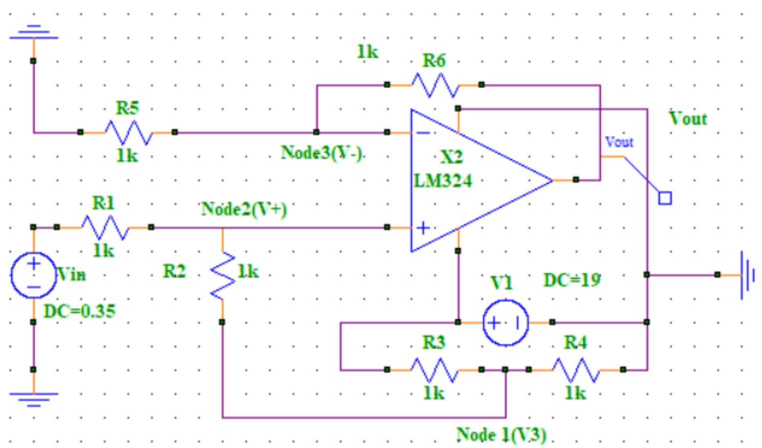

Figure 2: Candidate Design for Non-Inverting Operational Amplifier

This design is a modified version of the Non-Inverting Op-Amp. The basic Non-Inverting Op-Amp design fails to meet the engineering specification (as discussed in the research). In order to meet the 0.25 positive offset goal, a voltage divider design is used with a 19V voltage source to change the offset value. The function of the design is to amplify the input signal with a positive gain and offset.

$$\text{Vout} = \text{Vin}\times\frac{3.3-0.25}{0.35} + 0.25$$

To gain the expression above of V_{OUT} and V_{IN} , we can do nodal analysis. See Appendix A for the detailed steps of Nodal Analysis.

Based on the equation of V_{out} and V_{in}, we should see a line in the V_{OUT} versus V_{IN} plot with a positive offset. To validate this, A DC voltage source sweep in SuperSpice is used. V_{IN} is swept from 0 V to 0.35 V as this is the requirement with the Step = 0.01V.

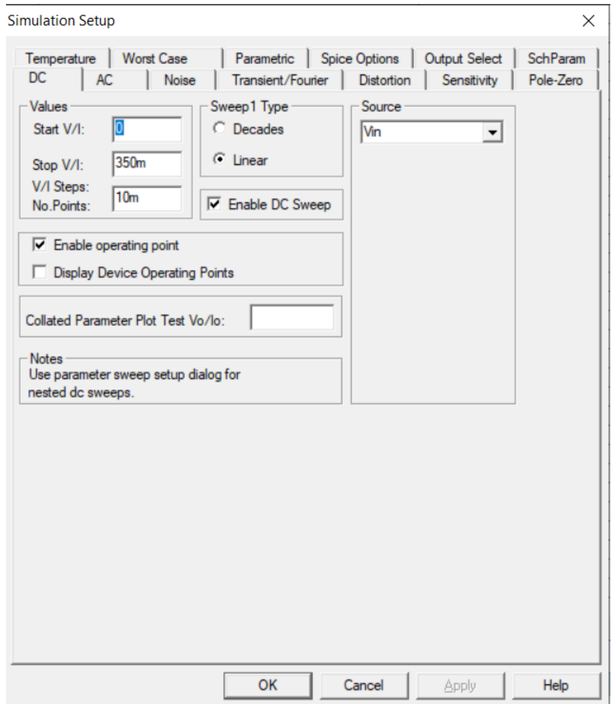

Figure 3 : The set-up of the DC voltage source sweep in SuperSpice

Based on the plot generated by *SuperSpice*, we can see a linear relationship between V_{OUT} and V_{IN} with a positive offset. This validates the design and we can just adjust the value of the resistance to get the expression we want.

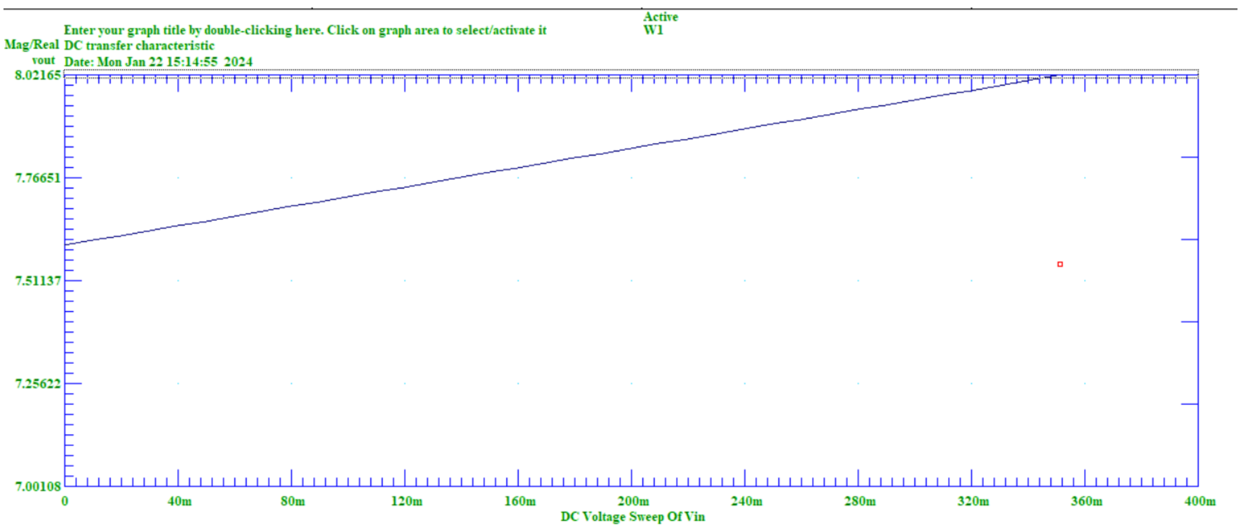

Figure 4: V_{OUT} versus V_{IN} Plot

2.2 Candidate Design 2: Difference Operational-Amplifier

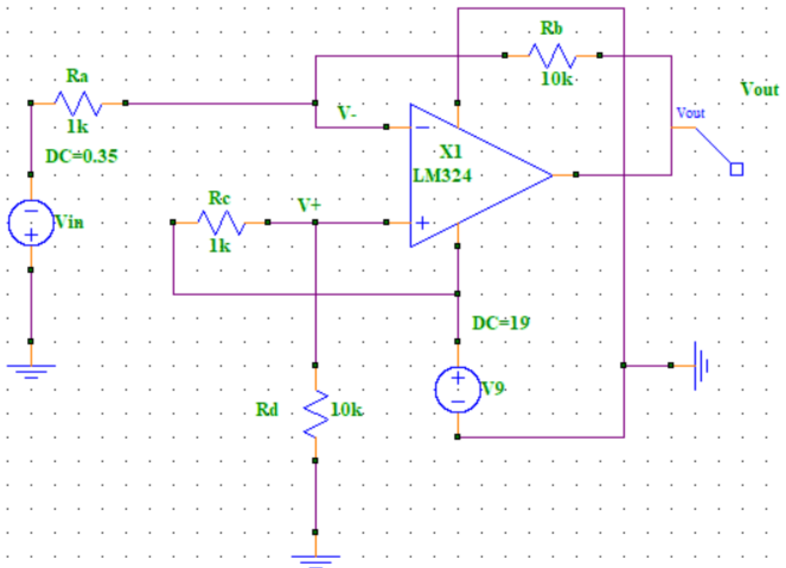

Figure 5 : Candidate Design for Difference Operational-Amplifier

To get the expression between V_{OUT} and V_{IN} of this candidate design, we use nodal analysis. See Appendix B for the analysis.

To validate this circuit, a DC voltage sweep in *SuperSpice* is performed with the similar settings to the previous one [Figure 3].

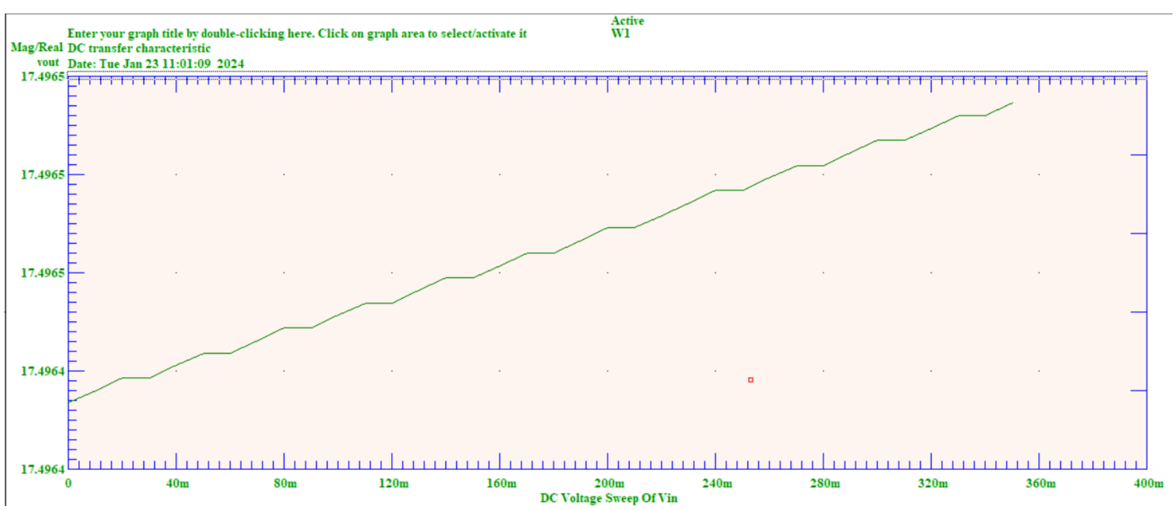

Figure 6: V_{OUT} versus V_{IN} plot

Based on Figure 6, we can see that the slope of V_{OUT} and V_{IN} is positive with a positive offset, but their relationship is not perfectly linear. The reason for a non-linear relationship is assumed to be due to the use of the step function instead of the continuous function to plot the relationship. Moreover, this cannot work due to the fact that V_- should be zero, as this operational amplifier does not allow negative voltages to enter the V_- terminal. Further details will be discussed in Section 3.0.

3.0 Final Candidate

Upon comparing the two candidate designs, the team decided to create the final design based on the first candidate design. It was chosen because its simulation result gave a perfectly linear relationship and positive offsets. The reason behind not selecting the second design is that it contains a negative voltage directly entering the "-" terminal of the op-amp, which is not physically practical for the particular op-amp we are using (LM324N). Based on Candidate Design 1, and after conducted continuous trials to find the appropriate resistance value, the final schematic is the following:

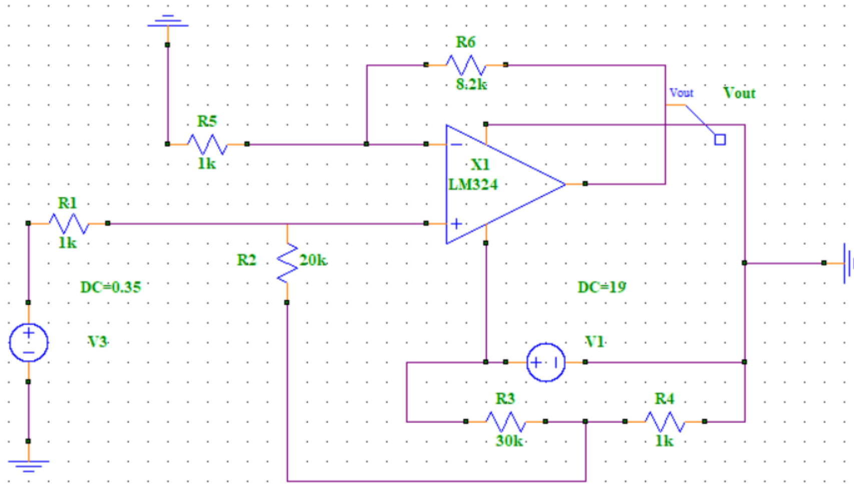

Figure 7: Final Circuit Design for Power Supply

Where R1 = 1k Ω, R2 = 20K Ω , R3 = 30K Ω, R4 = 1k Ω, R5 = 1kΩ and R6 = 8.2K Ω

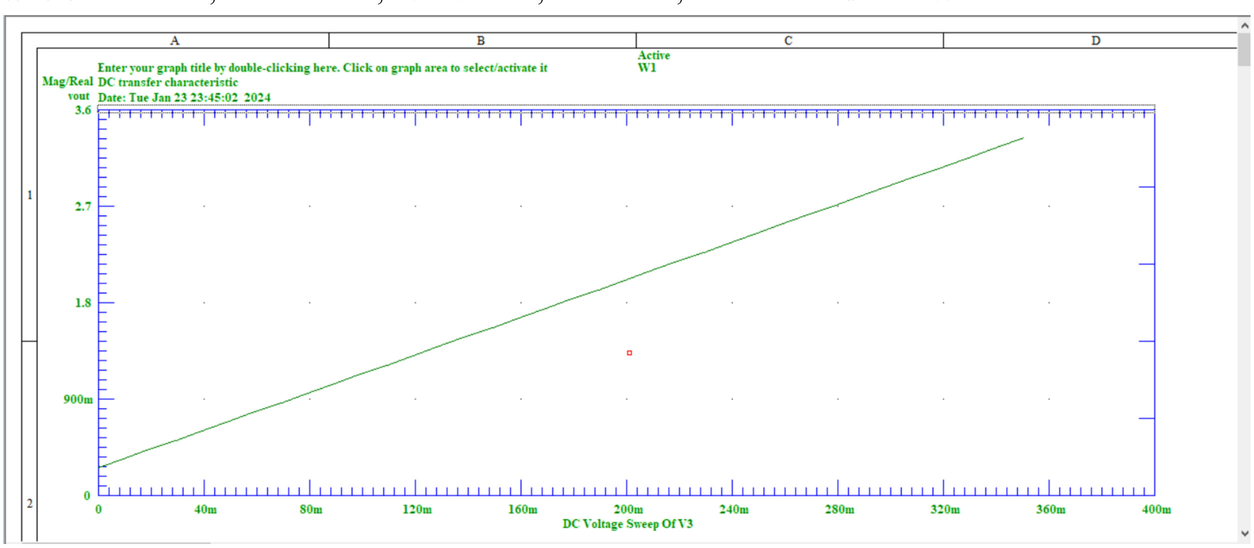

Figure 8: V_{OUT} versus V_{IN} for the final design

Based on Figure 8, it is clear to see V_{OUT} is proportional to V_{IN} with a positive offset.

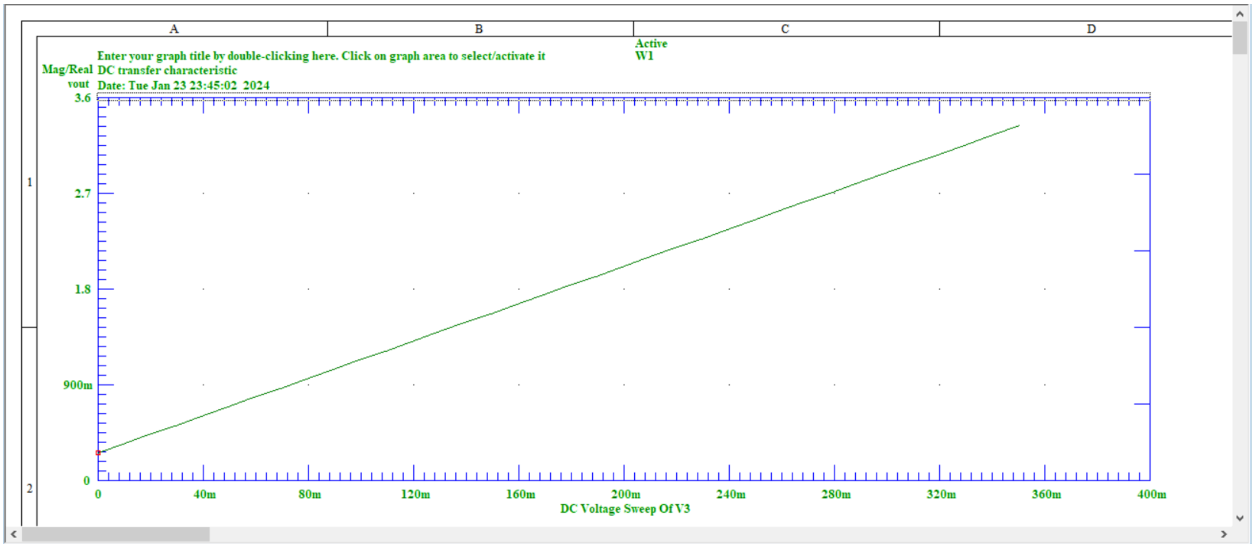

Figure 9: V_{OUT} at V_{IN} =0

Based on Figure 9, V_{OUT} = 0.262034 when V_{IN} = 0V. This value is within the 5% error range of 0.25, which is 0.2625V. Furthermore, V_{OUT} = 3.33497V when V_{IN} = 0.35 V, which is also within the 5% error range of 3.3, which is 3.465 V.

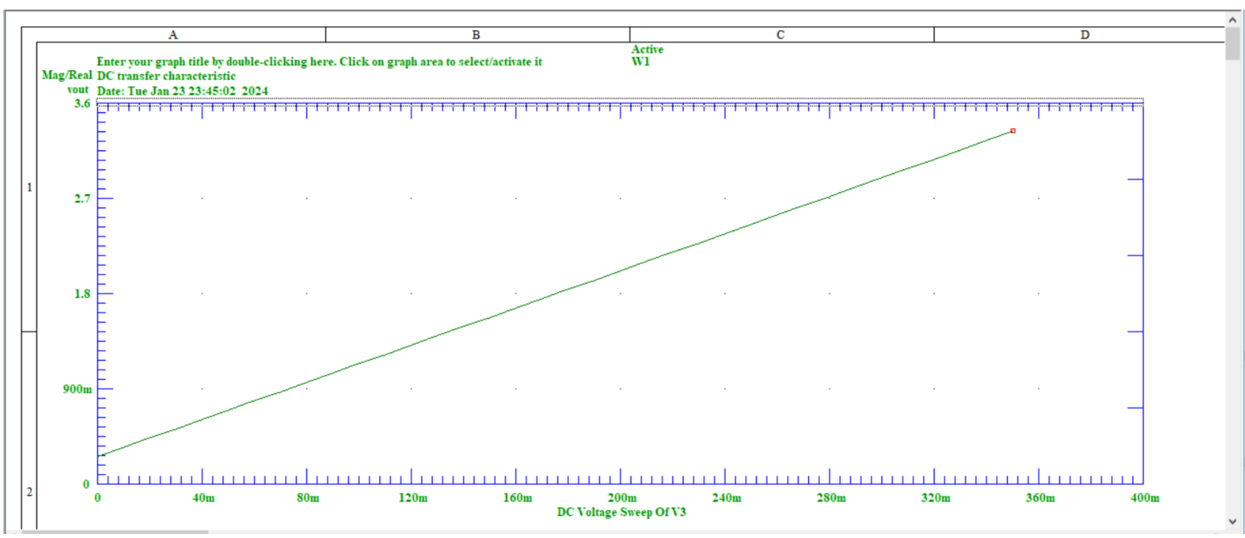

Figure 10: V_{OUT} at V_{IN} = 0.35V

In summary, the proposed final design successfully meets the engineering specifications laid out in the assignment. This is the design that will be transformed into a PCB in the upcoming Design Assignments, and it accomplishes the task of amplifying the input signal with a positive gain and offset.

[4.0] References:

[1] *W. Storr, "Non-inverting operational amplifier - the non-inverting op-amp," Basic Electronics Tutorials,* *https://www.electronics-tutorials.ws/opamp/opamp_3.html* *(accessed Jan. 23, 2024).*

[2] *W. Storr, "Differential amplifier - the voltage subtractor," Basic Electronics Tutorials,* *https://www.electronics-tutorials.ws/opamp/opamp_5.html* *(accessed Jan. 23, 2024).*

[5.0] Appendices:

Appendix A:

Node 1:

$$\frac{V_3 - 19}{R3} + \frac{V_3}{R4} + \frac{V_3 - V_+}{R2} = 0$$

Multiplying R2×R3×R4 to both sides

$$V_3\,(R2\times R3 + R2\times R4 + R3\times R4) = 19\times R2\times R4 + R3\times R4\times V_+$$

Now, assigning (R2×R3 + R2×R4 + R3×R4) as A for simplicity.

$$V_3 = \frac{1}{A}\,(19\times R2\times R4 + R3\times R4\times V_+)$$

Node 2:

$$\frac{V_+ - V_{IN}}{R1} + \frac{V_+ - V_3}{R2} = 0$$

Assuming an ideal op-amp (no current will flow in the V_+ and V_- terminals). Multiply R1×R2 to both sides and simplify, we can get:

$$V_+ = \frac{R2}{R2+R1}\times V_{IN} + \frac{R1}{R2+R1}\times V_3.$$

$$\text{As } V_3 = \frac{1}{A}\,(19\times R2\times R4 + R3\times R4\times V_+),$$

$$V_+ = \frac{R2}{R2+R1}\times V_{IN} + \frac{R1}{R2+R1}\times \frac{1}{A}\,(19\times R2\times R4 + R3\times R4\times V_+).$$

Rearrange the expression and we get

$$V_+\left(1 - \frac{R1}{R2+R1}\times \frac{1}{A}\times R3\times R4\right) = \frac{R2}{R2+R1}\times V_{IN} + \frac{R1}{R2+R1}\times \frac{1}{A}\,19\times R2\times R4.$$

Similarly, we can assign $\left(1 - \frac{R1}{R2+R1}\times \frac{1}{A}\times R3\times R4\right)$ as B thus:

$$V_+\times B = \frac{R2}{R2+R1}\times V_{IN} + \frac{R1}{R2+R1}\times \frac{1}{A}\,19\times R2\times R4.$$

$$V_+ = \frac{1}{B}\times \frac{R2}{R2+R1}\times V_{IN} + \frac{R1}{R2+R1}\times \frac{1}{A\times B}\,19\times R2\times R4$$

Node 3: (Assuming ideal op-amp, $V_- = V_+$)

$$\frac{V_+}{R5} + \frac{V_+ - V_{out}}{R6} = 0$$

multiply R5×R6 to both sides and simplify,

$$V_+ = \frac{R5}{R6+R5} \times V_{out}.$$

$$V_+ = \frac{1}{B} \times \frac{R2}{R2+R1} \times V_{IN} + \frac{R1}{R2+R1} \times \frac{1}{A \times B} 19 \times R2 \times R4,$$

$$\frac{R5}{R6+R5} \times V_{out} = \frac{1}{B} \times \frac{R2}{R2+R1} \times V_{IN} + \frac{R1}{R2+R1} \times \frac{1}{A \times B} 19 \times R2 \times R4,$$

Assign $\frac{R5}{R6+R5}$ as C, making the overall expression between V_{OUT} and V_{IN}:

$$V_{out} = \frac{1}{B \times C} \times \frac{R2}{R2+R1} \times V_{IN} + \frac{R1}{R2+R1} \times \frac{1}{A \times B \times C} 19 \times R2 \times R4.$$

Compared this expression with the one we want:

$$V_{OUT} = V_{IN} \times \frac{3.3 - 0.25}{0.35} + 0.25, \text{ we can let}$$

$$\frac{1}{B \times C} \times \frac{R2}{R2+R1} = \frac{3.3 - 0.25}{0.35} \text{ and } \frac{R1}{R2+R1} \times \frac{1}{A \times B \times C} 19 \times R2 \times R4 = 0.25.$$

Appendix B:

Node V_+

Using voltage division rule, $V_+ = \frac{Rd}{Rc+Rd} \times 19$, since an ideal op-amp, $V_+ = V_-$

Node V_-

$$\frac{V_+ - V_{IN}}{Ra} + \frac{V_+ - V_{out}}{Rb} = 0$$

Multiply Ra×Rb to both sides

$$V_+ \times (Ra + Rb) = Ra \times V_{OUT} - Rb \times V_{IN}$$

Substitute $V_+ = \frac{Rd}{Rc+Rd} \times 19$

$$\frac{Rd}{Rc+Rd} \times 19 \times (Ra + Rb) = Ra \times V_{OUT} - Rb \times V_{IN}$$

Rearrange the expression, $V_{OUT} = \frac{Rd}{Ra \times (Rc+Rd)} \times (Ra+Rb) \times 19 - \frac{Rb}{Ra} \times V_{IN}$ since the voltage source of V_{IN} is reversed:

$$V_{OUT} = \frac{Rd}{Ra \times (Rc+Rd)} \times (Ra+Rb) \times 19 + \frac{Rb}{Ra} \times V_{IN}.$$

Compared this expression to the expression we want : $V_{OUT} = V_{IN} \times \frac{3.3 - 0.25}{0.35} + 0.25$, we can achieve by setting:

$$\frac{Rb}{Ra} = \frac{3.3 - 0.25}{0.35} \text{ and } \frac{Rd}{Ra \times (Rc + Rd)} \times (Ra+Rb) \times 19 = 0.25.$$

1.0 Design Features and Updated Schematic

Figure 1: Schematic of Design Assignment 1's Circuit with the New Filter Design

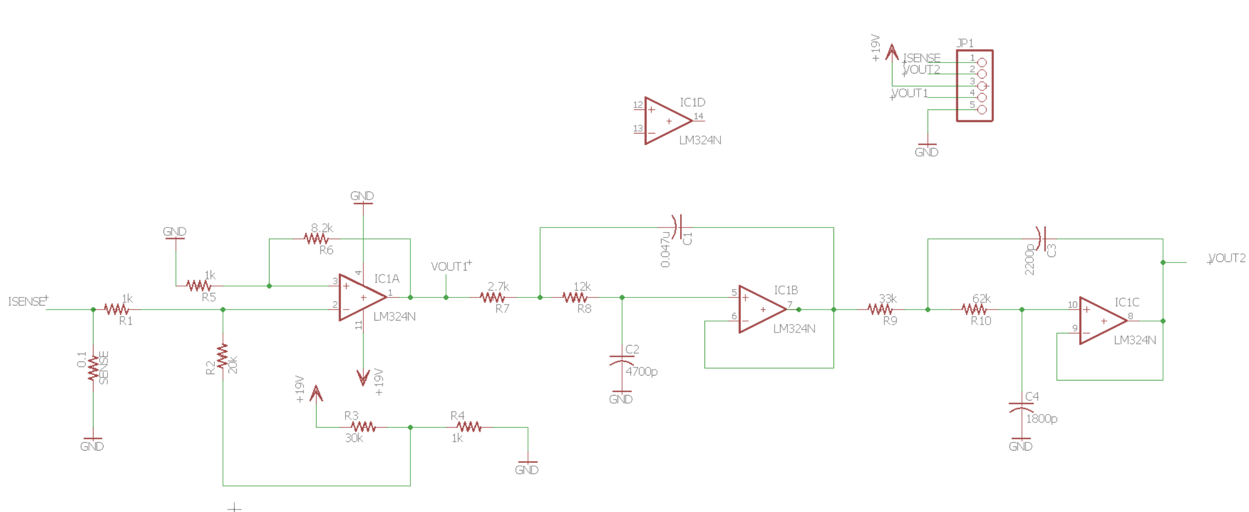

Figure 1: Schematic of Design Assignment 1's Circuit with the New Filter Design

<u>Changes from Design Assignment 1:</u>

- A 0.1 Ω sense resistor connected to the ground is added.
- Two low pass filters are added to the design.

The details of the filter design are further explained below:

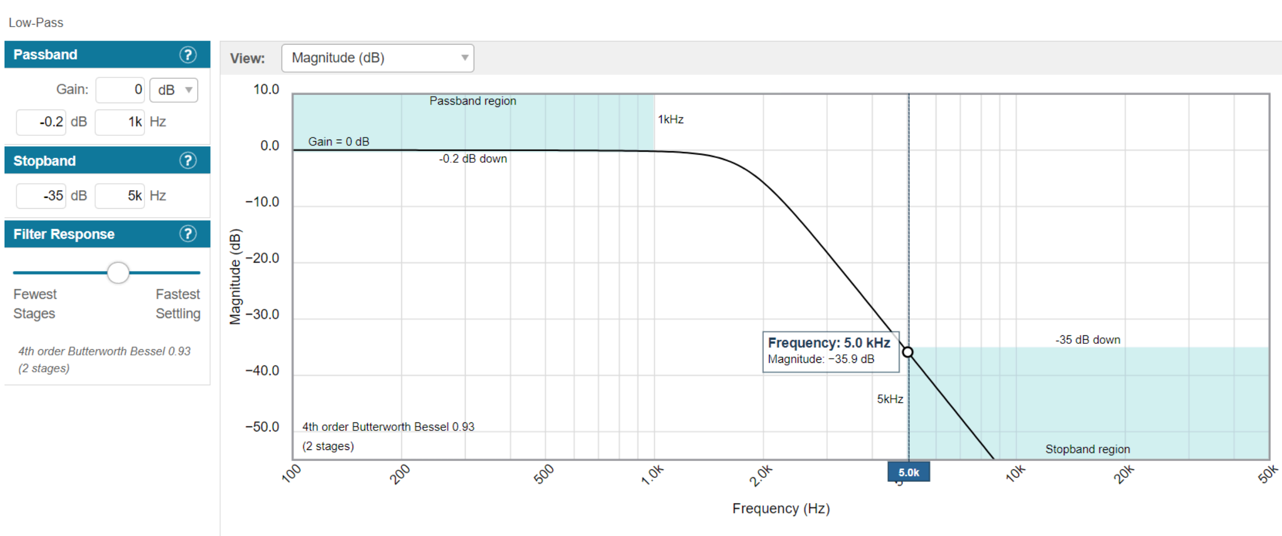

Figure 2: Graph of Passband and Stopband Regions with Frequency Response

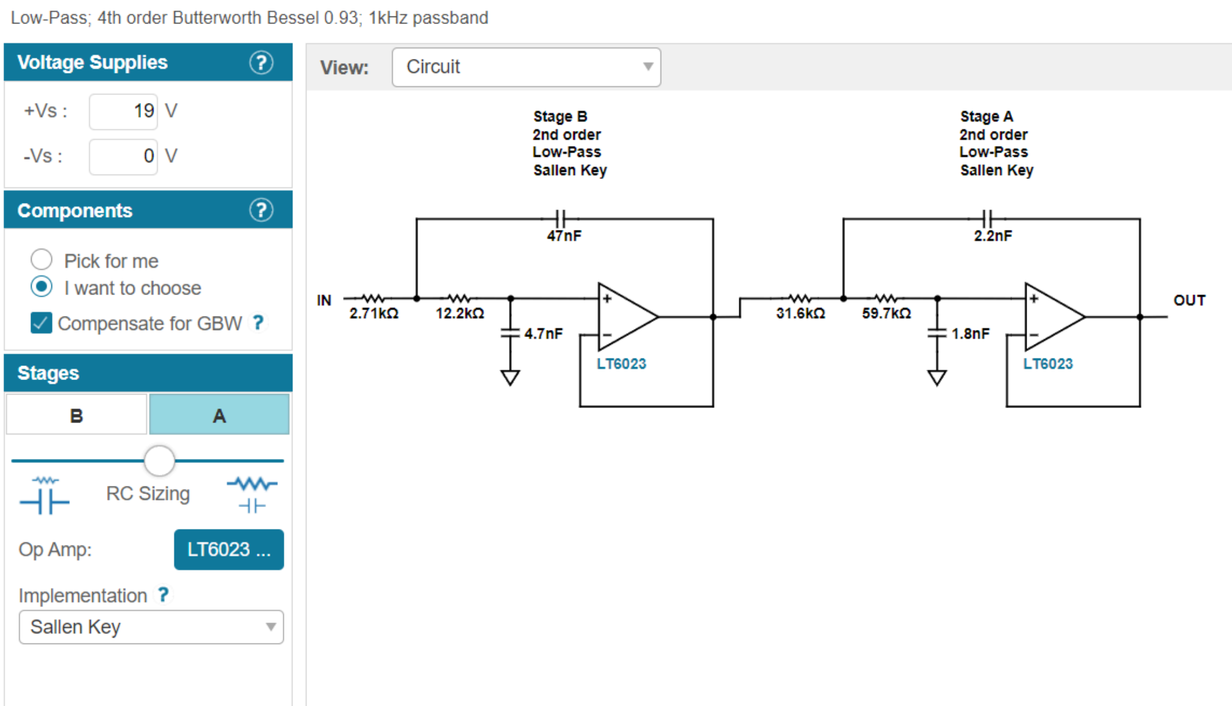

Figure 3: Filter Design created using Analog Devices Tool

Figures 2 and 3 showcase how Analog Devices' Filter Tool was used to create a filter that met the desired conditions outlined in the report instructions. The final design choices were as follows:

- Two Op-Amps (Stages)
- 4th Order Buttersworth Bessel 0.93, 1kHz Passband
- 0 dB Gain
- Cut-off Frequency of 1 kHz
- Passband Attenuation of -0.2 dB
- Stopband Frequency of 5 kHz
- Stopband Attenuation of 35.9 dB

2.0 Bill of Materials (MIE346 DA2 Component Order)

Sweeping through the RC Sizing slider pictured in Figure 3 for both Stages A and B was used to find component values that followed the charts in the Design Assignment 2 instructions. Those were the following:

Stage A:
Resistors: The selection of resistor is based on the 10% error of the target resistance

- 31.6 kΩ -> 33 kΩ (Chart Value) (R9)
- 59.7 kΩ -> 62 kΩ (Chart Value)(R10)

Capacitors:

- 1.8 nF -> 1800 pF (Chart Value)(C4)
- 2.2 nF -> 2200 pF (Chart Value)(C3)

Stage B:
Resistors:

- 2.7 kΩ -> 2.7 kΩ (Chart Value)(R7)
- 12.2 kΩ -> 12 kΩ (Chart Value)(R8)

Capacitors:

- 47 nF -> 0.047 uF (Chart Value)(C1)
- 4.7 nF -> 4700pF (Chart Value)(C2)

Design Assignment 1 Circuit:
Resistors:

- 0.1 Ω (Sense)(Rsense)
- 1 kΩ (R1)
- 1 kΩ (R4)
- 1 kΩ (R5)
- 8.2 kΩ (R6)
- 20 kΩ (R2)
- 30 kΩ (R3)

The above chart values have been added to the Bill of Materials Spreadsheet.

3.0 PCB Layout
The schematic created in Figure 1 was used to create a PCB layout. Three PCB layouts can be found below, a top, bottom and combined layer.

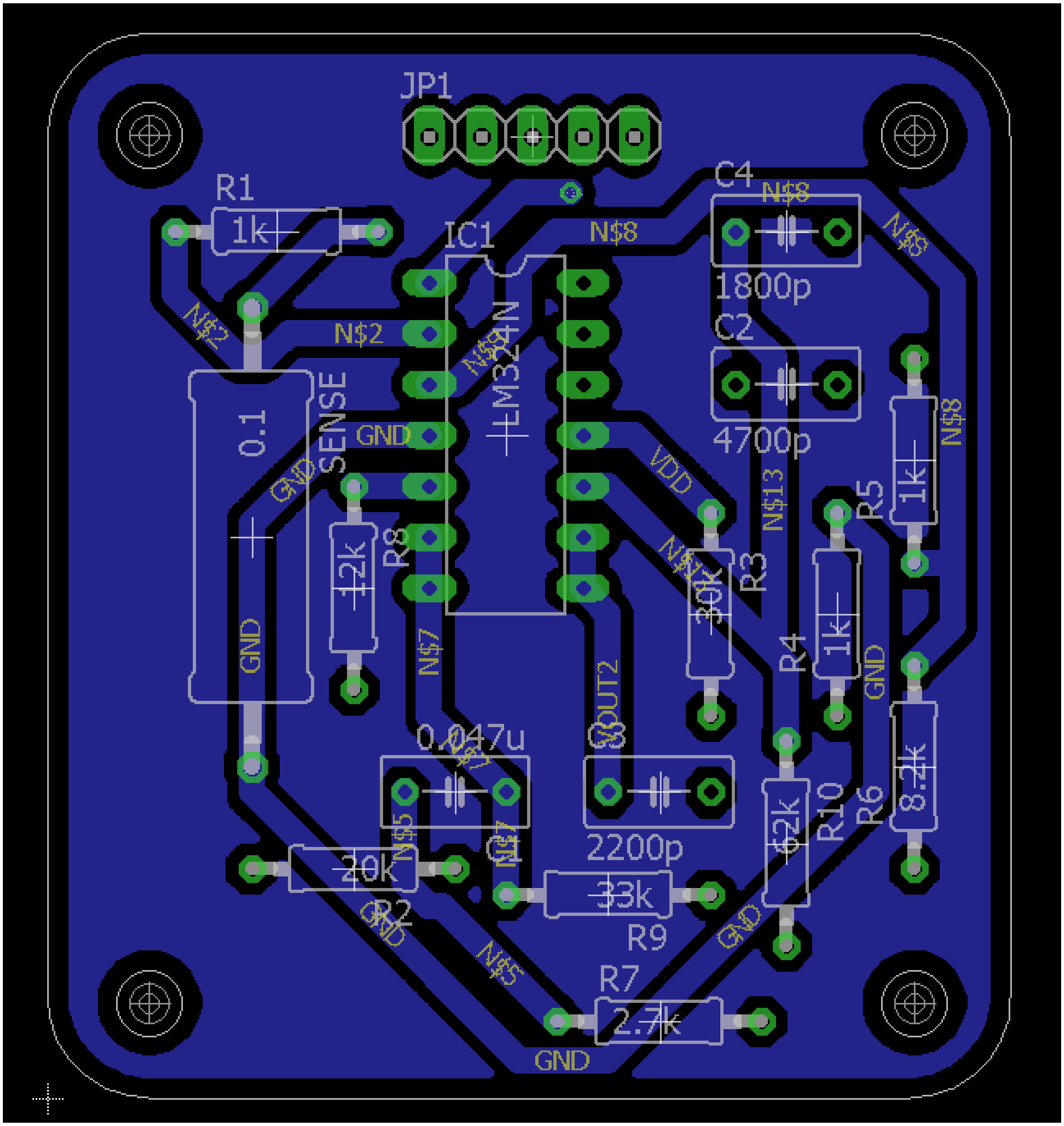

Figure 4: Top Component Layer PCB Layout in Eagle

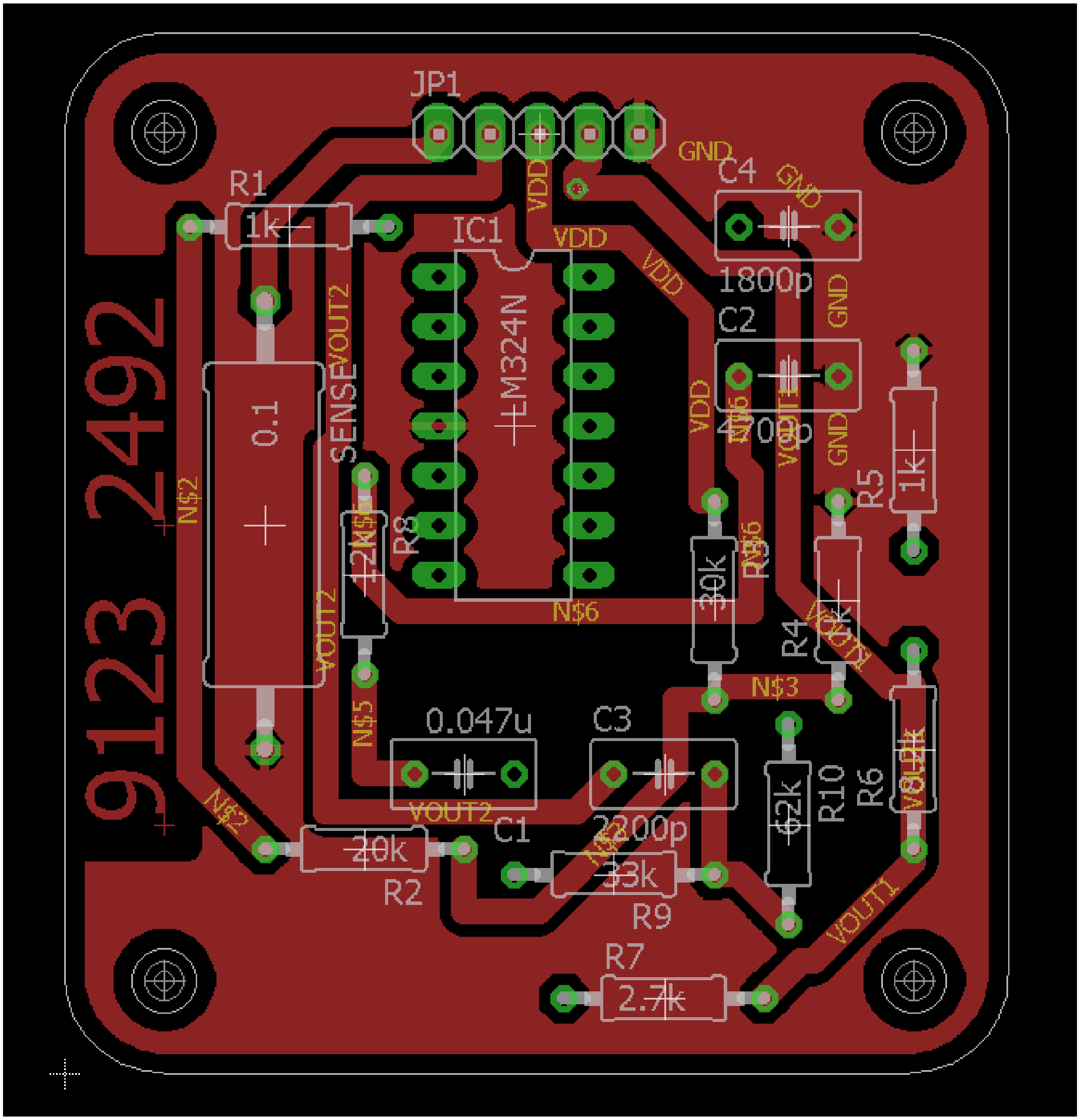

Figure 5: Bottom Component Layer PCB Layout in Eagle

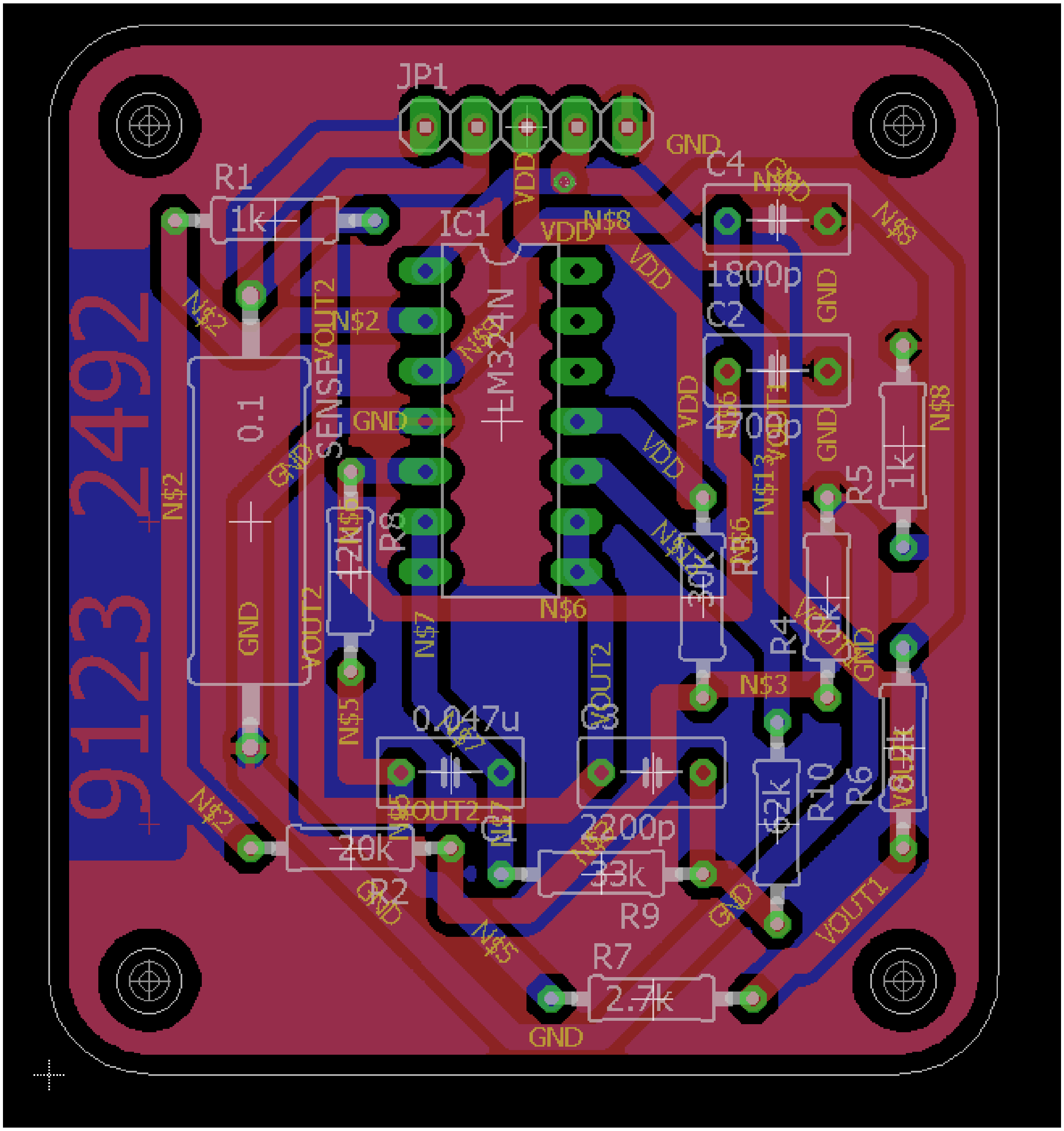

Figure 6: Combined Component Layer PCB Layout in Eagle

Note: Due to an unforeseen issue regarding our circuit board from Design Assignment 2, a reference board needed to be used for this report.

Reason for Failure: Our original design had the first operational amplifier flipped in the wrong direction (i.e. the + and - were swapped). The sign was correct in the design assignment 1 but was swapped by mistakes in the design assignment 2. As this error was missed, our PCB was printed in the wrong way. The error was only caught in the late stages of the assignment, thus the reference board was necessary for the remainder of the assignment.

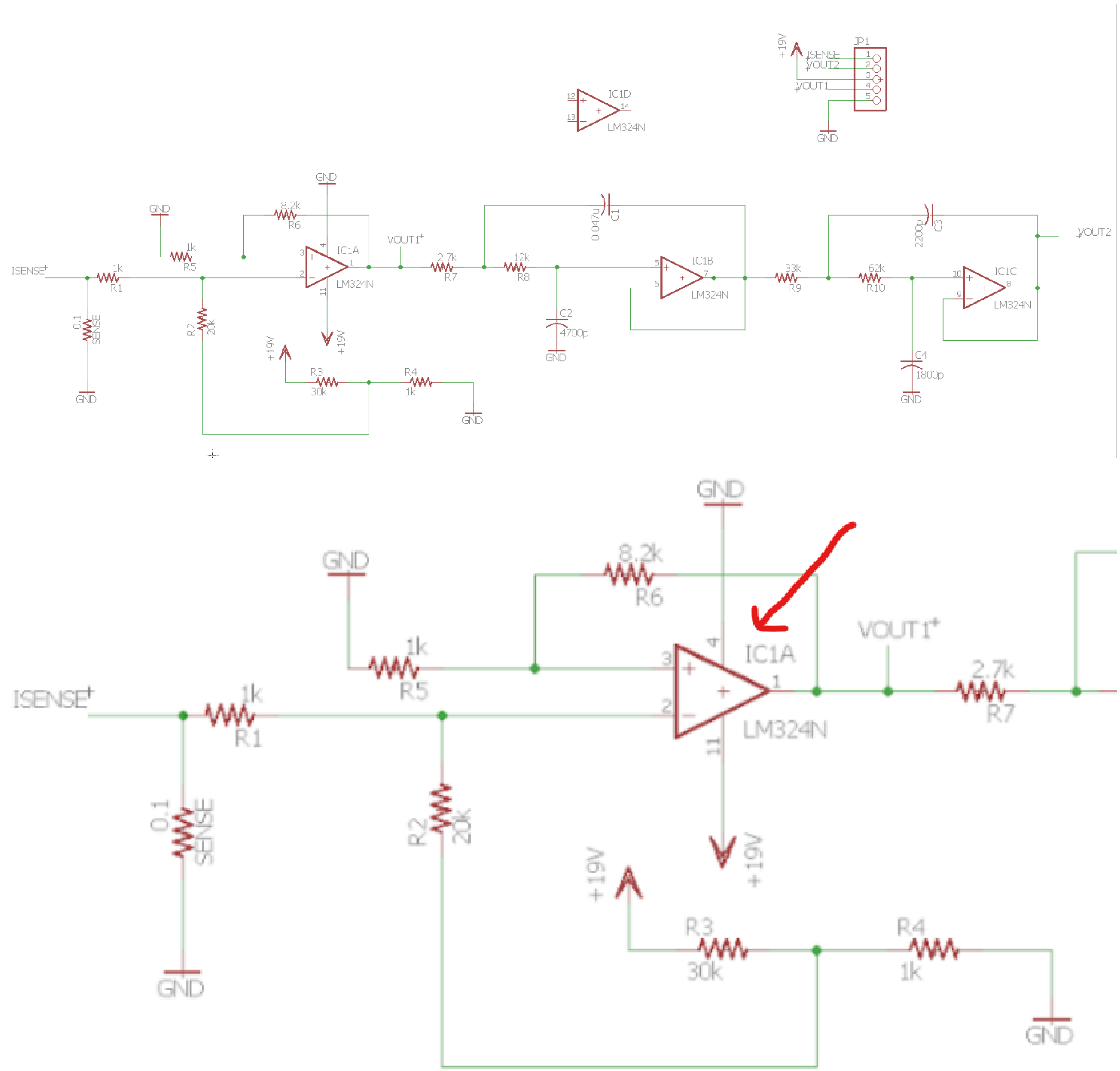

Figure 1: Original Design Schematic + Zoom-In on Error

1.0 Table of Measured Output Voltages With Respect To I_{SENSE}

I_{sense} (A)	V_{OUT1} (Unfiltered Output)	V_{OUT2} (Filtered Output)	Expected V_{OUT}(V)
0.000	0.251	0.252	0.25
0.010	0.262	0.260	0.26
0.020	0.274	0.272	0.27
0.030	0.284	0.283	0.28
0.050	0.301	0.300	0.29
0.100	0.352	0.350	0.34
0.150	0.400	0.398	0.38
0.200	0.450	0.449	0.42
0.300	0.547	0.545	0.51
0.400	0.623	0.622	0.60
0.500	0.759	0.754	0.69
1.000	1.310	1.300	1.12
2.000	1.950	1.930	1.99
2.500	2.470	2.440	2.43
3.000	3.000	2.940	2.86
3.200	3.500	3.400	3.032

Table 1. The Filtered voltage Output and Unfiltered Voltage Output with respect to the Sense Current

1.1 Expectations

Analysis of the above table makes it safe to assume that the circuit worked correctly. This is because, both the unfiltered and filtered outputs closely matched each other, which is what should happen in the given scenario. Moreover, they closely resemble the expected V_{out} values, which shows that the designed PCB matches the expectations. There are minor discrepancies as higher I_{sense} values are reached, but these can be attributed to minor errors related to noise, equipment defects (such as the oscilloscope probe), and measurement errors. The main reason behind these discrepancies could be that the gain is not perfectly matched, due to the limited choice of the resistor and capacitor values when designing the circuit.

2.0 Comparison of the Filtered and Unfiltered Output with a Step Input

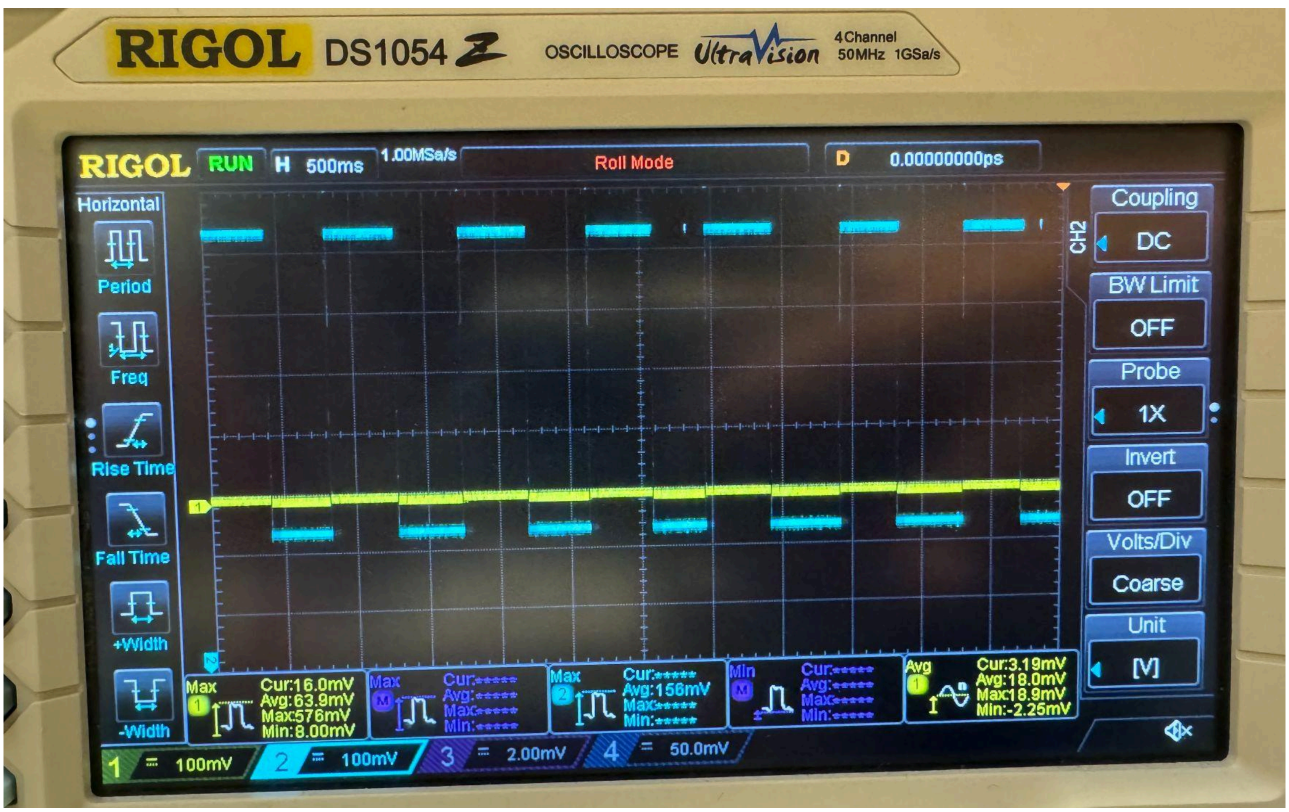

Figure 2: Comparison of the Unfiltered Voltage Output (in Blue) and Filtered Voltage Output (in Yellow)

2.1 Justification

Figure 2 shows a comparison of the unfiltered and filtered voltage outputs. As we can see, there are no unusual outcomes. The reason to use low-pass filters is to reduce noise and remove high-frequency signals (the manually produced step functions in our case) thus making the output consistent and smooth. The blue lines (unfiltered) are easily affected by the step response (high frequency signals) and noise, but the yellow lines (filtered) are not. Hence, the purpose of the filter was served. It created a consistent output voltage, just like it was expected to.

3.0 Images of the Soldered PCB

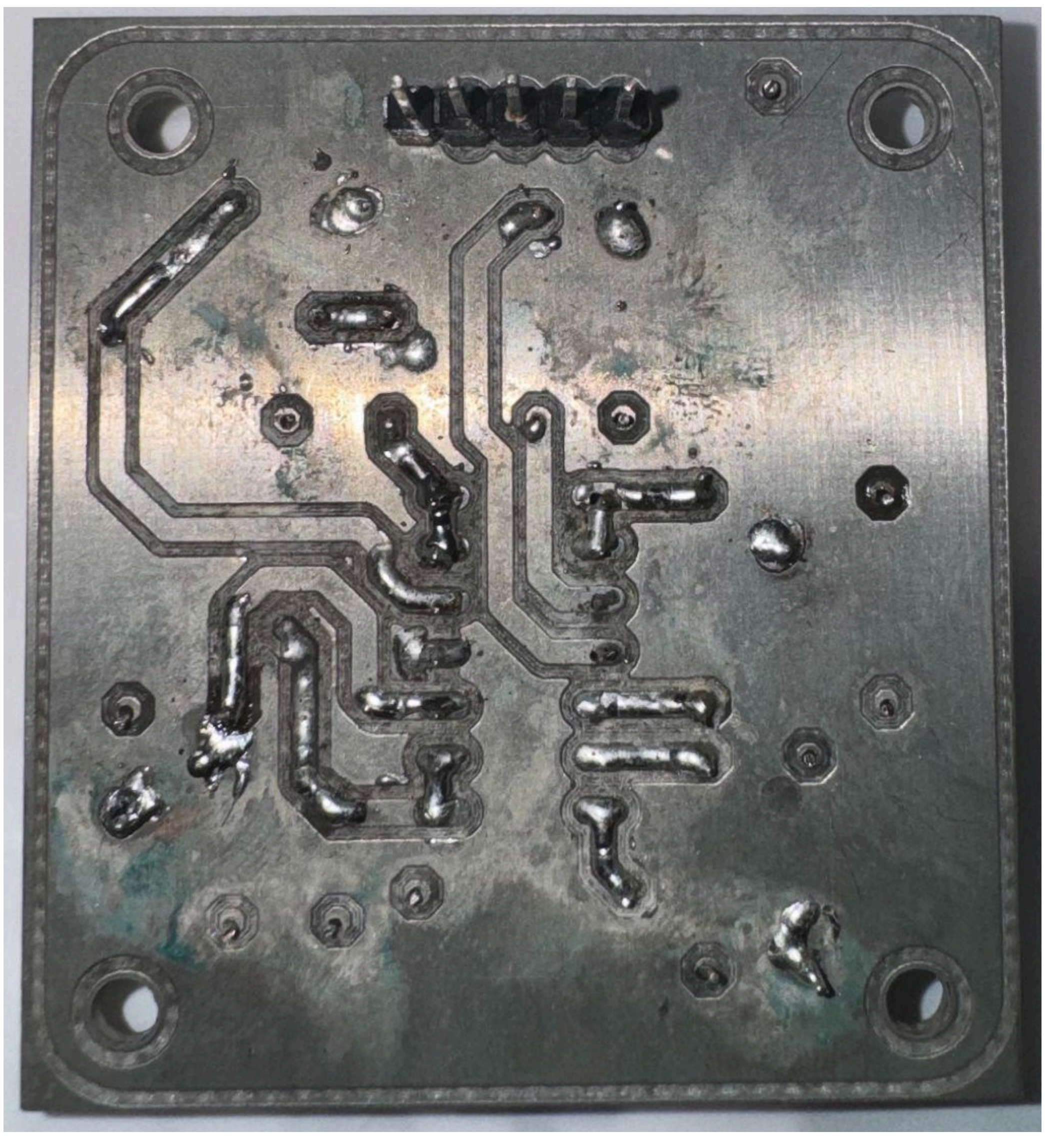

Figure 3.1: The Bottom Layer of the Soldered PCB

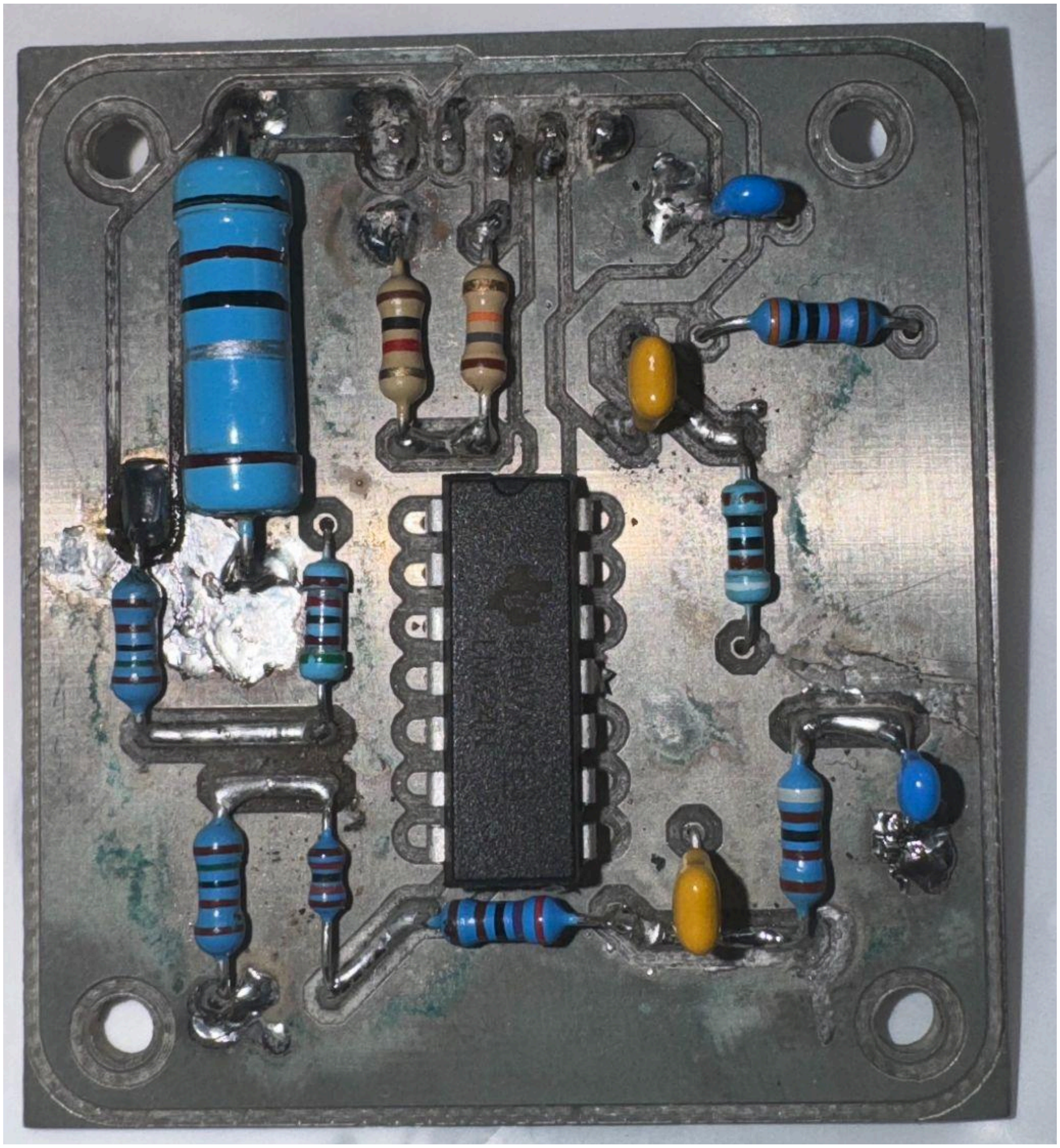

Figure 3.2: The Top Layer of the Soldered PCB

Deliverable 1:

I_{SENSE} [A]	I_{SENSE} (Direct/Unfiltered)	I_{SENSE} (Filtered)
0.010	272	298
0.020	280	307
0.030	289	318
0.050	307	337
0.100	353	387
0.150	402	437
0.200	447	486
0.300	538	591
0.400	629	684
0.500	719	780
1.000	1180	1260
2.000	2096	2334
2.500	2555	2807
3.000	3019	3284
3.200	3023	3478

Table 1: Comparison of the Two ADC Values for the Unfiltered and Filtered Outputs

Upon analyzing the table, we can see that there are not any significant discrepancies or errors in the data. When measuring any data, there is a margin of error associated with calibration and measurement inefficiencies, but despite this, the data in the table shows a linear relationship, which is exactly what we expect.

Deliverable 2:

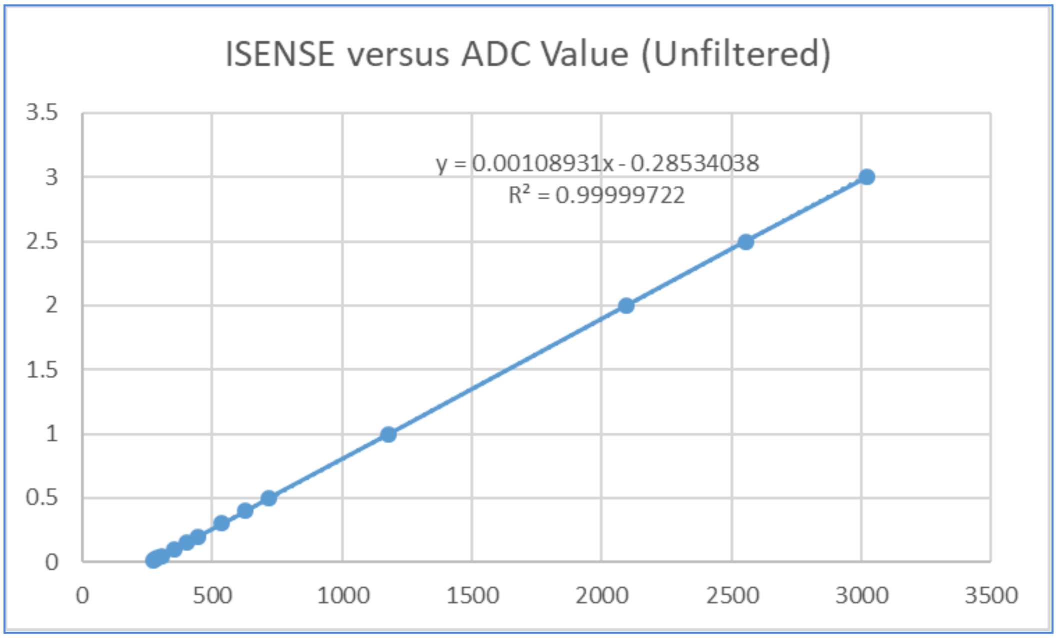

Figure 1: Graph of the Unfiltered ISENSE Versus ADC Values

Note: This plot deletes the data point for 3.2 A for better accuracy and linearity.
Based on Figure 1, the equation of the trendline for the unfiltered fitted line is:

$$y = 0.00108931x - 0.28534038$$

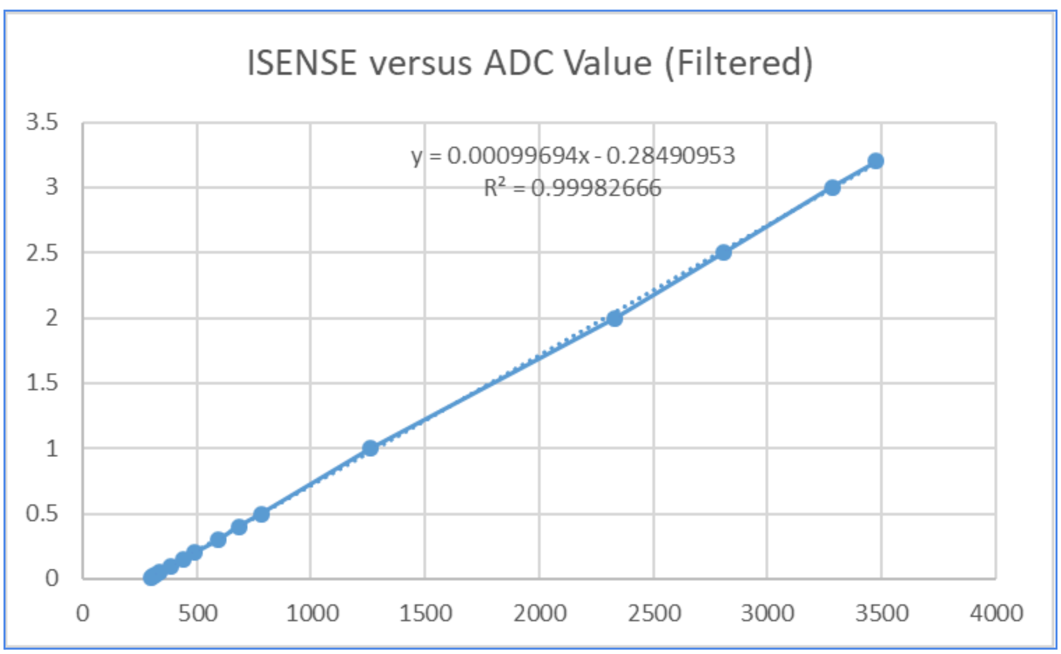

Figure 2: Graph of the Filtered ISENSE Versus ADC Values

Based on Figure 2, the equation of the trendline for the filtered fitted line is:

$$y = 0.00099694x - 0.28490953$$

Even the linearity of the unfiltered plot (R^2 = 0.99999722) is slightly better than that of the filtered plot (R^2 = 0.99982666). The filter plot is chosen for the remainder of the design assignment as it does not delete any data point.

Deliverable 3A:

Trial	V_{out} (Target) [V]	V_{out} (Measured) [V]
1	0.01	0.13
2	0.5	0.59
3	1	1.09
4	3	3.08
5	5	5.08
6	7	7.08
7	9.35	9.42
8	14.47	14.54
9	15	15.07

Table 2: Comparison of V_{out} from the UI Versus Measured V_{out}

Comment: Based on Table 2, we can see the difference between V_{out} (Target) and V_{out} (Measured) is smaller when V_{out} (Target) is larger than 0.01V. This implies that there is an optimum range of the voltage that our design can produce. Also, we can see there is a consistent 0.07-0.09 V offset between V_{out} (Target) and V_{out} (Measured) once Vout (Target) is larger than 0.01V. This offset can be due to the limited choice of capacitor and resistor values so we cannot get the perfect gain value for the design.

Deliverable 3B:

Trial	I (Target) [A]	I (Measured) [A]
1	**0.01**	**0.006**
2	**0.61**	**0.621**
3	**1.15**	**1.189**
4	**1.5**	**1.556**
5	**1.6**	**1.665**
6	**1.73**	**1.800**
7	**2.2**	**2.21**
8	**2.5**	**2.52**
9	**3.1**	**3.13**

Table 3: Comparison of I_{out} from the UI Versus Measured I_{out}

Comment: The quality of results from both tables expertly aligns with what was expected from the power supply. A difference of approximately 0.8 V and 0.06 A is present between the target values and those measured, which was satisfactory enough to avoid recalibration. No discrepancy between the target and measured values exceeds a 4 percent margin of error, further showcasing how accurate and precise the results are.

Deliverable 3C:

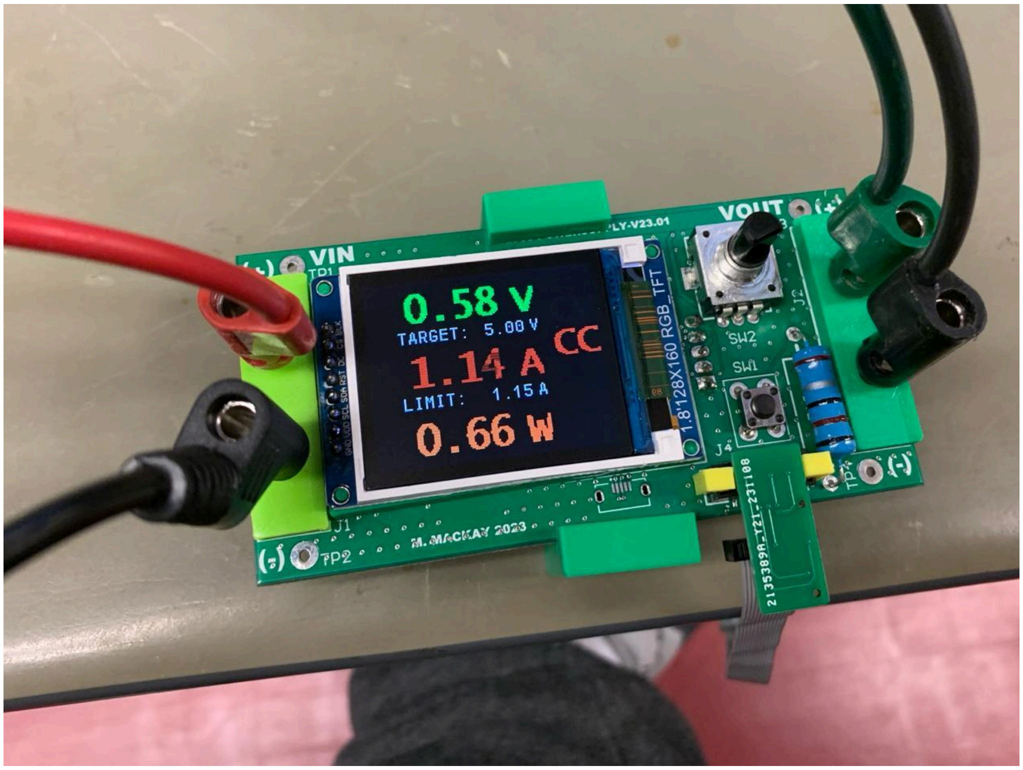

Figure 3: Working Power Supply with LCD Display Running Firmware

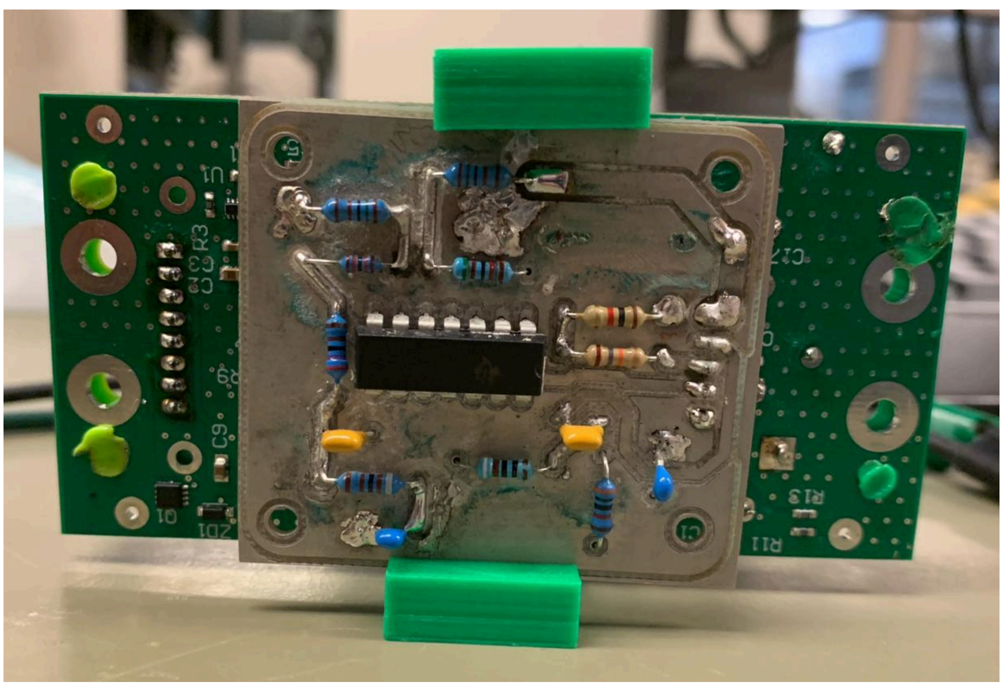

Figure 4: Backside of the Project Showcasing Mounted PCB